

TOP 50 DSA

Interview Questions

FOR MAANG





Disclaimer

Everyone learns at a different
pace and plan.

It is not the number of questions
that you solve but gaining the
ability to analyse, identify and
come up with solutions for related
problems is what matters.

Arrays



Question 1.

Given an array of integers and an integer target, return indices of the two numbers such that they add up to target.

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Question 2.

Given an integer array nums, find the subarray with the largest sum, and return its sum.

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Question 3.

Given an array nums with n objects colored red, white, or blue, sort them in-place so that objects of the same color are adjacent, with the colors in the order red, white, and blue.

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Question 4.

Given an array `nums` of `n` integers, return an array of all the unique quadruplets `[nums[a], nums[b], nums[c], nums[d]]` such that:

$0 \leq a, b, c, d < n$

`a`, `b`, `c`, and `d` are distinct.

`nums[a] + nums[b] + nums[c] + nums[d] == target`

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Question 5.

You are given an array of prices where `prices[i]` is the price of a given stock on an `i`th day. You want to maximise your profit by choosing a single day to buy one stock and choosing a different day in the future to sell that stock. Return the maximum profit you can achieve from this transaction. If you cannot achieve any profit, return 0.

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Strings



Question 1.

Write a function that reverses a string. The input string is given as an array of characters `s`. You must do this by modifying the input array in-place with $O(1)$ extra memory.

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Question 2.

Given a string `s`, sort it in decreasing order based on the frequency of the characters. The frequency of a character is the number of times it appears in the string. Return the sorted string. If there are multiple answers, return any of them.

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Question 3.

Given two strings s1 and s2, return true if s2 contains a permutation of s1, or false otherwise.

In other words, return true if one of s1's permutations is the substring of s2.

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Question 4.

Given a string s, partition s such that every Substring of the partition is a Palindrome. Return all possible palindrome partitioning of s.

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Question 5.

You are given a string s and an integer k. You can choose any character of the string and change it to any other uppercase English character. You can perform this operation at most k times.

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Recursion



Question 1.

Given the head of a linked list and an integer val, remove all the nodes of the linked list that has `Node.val == val`, and return the new head.

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Question 2.

Given the head of a singly linked list, reverse the list, and return the reversed list.

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Asked in :   



Question 3.

Given an integer array nums of unique elements, return all possible

Subsets (the power set).

The solution set must not contain duplicate subsets. Return the solution in any order

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Asked in :  

Question 4.

Given n pairs of parentheses, write a function to generate all combinations of well-formed parentheses.

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Asked in :    

Hashing



Question 1.

Design a data structure that follows the constraints of a Least Recently Used (LRU) cache.

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Asked in :



intuit

Question 2.

Given an unsorted integer array nums, return the smallest missing positive integer.

You must implement an algorithm that runs in $O(n)$ time and uses constant extra space.

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Asked in :



Matrices

Question 1.

Given an $m \times n$ matrix, return all elements of the matrix in spiral order.

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Asked in :



Question 2.

Determine if a 9×9 Sudoku board is valid. Only the filled cells need to be validated according to the following rules:
Each row must contain the digits 1-9 without repetition.
Each column must contain the digits 1-9 without repetition.
Each of the nine 3×3 sub-boxes of the grid must contain the digits 1-9 without repetition.

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Asked in :



Question 3.

Given an $m \times n$ grid of characters board and a string word, return true if word exists in the grid.

The word can be constructed from letters of sequentially adjacent cells, where adjacent cells are horizontally or vertically neighboring. The same letter cell may not be used more than once.

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Asked in :



Linked List

Question 1.

Given the root of a binary tree, flatten the tree into a "linked list":

The "linked list" should use the same `TreeNode` class where the right child pointer points to the next node in the list and the left child pointer is always null.

The "linked list" should be in the same order as a pre-order traversal of the binary tree.

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Asked in :  

Question 2.

Given elements as nodes of the two linked lists. The task is to multiply these two linked lists, say L1 and L2.

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Question 3.

Given the head of a linked list, reverse the nodes of the list k at a time, and return the modified list. k is a positive integer and is less than or equal to the length of the linked list. If the number of nodes is not a multiple of k then left-out nodes, in the end, should remain as it is.

You may not alter the values in the list's nodes, only nodes themselves may be changed.

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Question 4.

You are given the heads of two sorted linked lists list1 and list2.

Merge the two lists in a one sorted list. The list should be made by splicing together the nodes of the first two lists. Return the head of the merged linked list.

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Question 5.

You are given the head of a singly linked-list. The list can be represented as:

$L_0 \rightarrow L_1 \rightarrow \dots \rightarrow L_{n-1} \rightarrow L_n$

Reorder the list to be on the following form:

$L_0 \rightarrow L_n \rightarrow L_1 \rightarrow L_{n-1} \rightarrow L_2 \rightarrow L_{n-2} \rightarrow \dots$

You may not modify the values in the list's nodes. Only nodes themselves may be changed.

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Question 6.

Given a linked list, swap every two adjacent nodes and return its head. You must solve the problem without modifying the values in the list's nodes (i.e., only nodes themselves may be changed.)

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Asked in :  



Bit Manipulation and Math

Question 1.

Given an array `nums` containing n distinct numbers in the range $[0, n]$, return the only number in the range that is missing from the array.

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Question 2.

Given an integer n , return an array `ans` of length $n + 1$ such that for each i ($0 \leq i \leq n$), `ans[i]` is the number of 1's in the binary representation of i .

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Asked in :  



Stacks and Queues



Question 1.

Given an array of integers heights representing the histogram's bar height where the width of each bar is 1, return the area of the largest rectangle in the histogram.

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Question 2.

Given a string s containing just the characters '(', ')', '{', '}', '[' and ']', determine if the input string is valid.

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Question 3.

Implement a last-in-first-out (LIFO) stack using only two queues. The implemented stack should support all the functions of a normal stack (push, top, pop, and empty).

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Question 4.

Implement the BSTIterator class that represents an iterator over the in-order traversal of a binary search tree (BST)

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Asked in :    

Question 5.

Given n non-negative integers representing an elevation map where the width of each bar is 1, compute how much water it can trap after raining.

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Asked in :   



Trees and Binary Search Trees

Question 1.

Given a binary tree, determine if it is height-balanced

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Question 2.

Given a binary tree, find the lowest common ancestor (LCA) of two given nodes in the tree.

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Asked in :



Question 3.

Given the root of a binary search tree, and an integer k, return the kth smallest value (1-indexed) of all the values of the nodes in the tree.

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Asked in :



Question 4.

Given the root of a binary tree, return the level order traversal of its nodes' values. (i.e., from left to right, level by level).

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Asked in :



Question 5.

You are given the root of a binary tree containing digits from 0 to 9 only.

Each root-to-leaf path in the tree represents a number.

For example, the root-to-leaf path $1 \rightarrow 2 \rightarrow 3$ represents the number 123. Return the total sum of all root-to-leaf numbers. Test cases are generated so that the answer will fit in a 32-bit integer.

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Asked in :



Question 6.

A path in a binary tree is a sequence of nodes where each pair of adjacent nodes in the sequence has an edge connecting them. A node can only appear in the sequence at most once. Note that the path does not need to pass through the root.

The path sum of a path is the sum of the node's values in the path.

Given the root of a binary tree, return the maximum path sum of any non-empty path.

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Tries



Question 1.

A trie (pronounced as "try") or prefix tree is a tree data structure used to efficiently store and retrieve keys in a dataset of strings. There are various applications of this data structure, such as autocomplete and spellchecker. Implement the Trie.

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Asked in :



Question 2.

Given an array of strings strs, group the anagrams together. You can return the answer in any order.

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Asked in :



Heaps



Question 1.

You are given an array of k linked-lists lists, each linked-list is sorted in ascending order.

Merge all the linked-lists into one sorted linked-list and return it.

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Question 2.

The median is the middle value in an ordered integer list. If the size of the list is even, there is no middle value, and the median is the mean of the two middle values.

Implement the MedianFinder class:

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Graphs

Question 1.

Given an $m \times n$ binary matrix `mat`, return the distance of the nearest 0 for each cell.

The distance between two adjacent cells is 1.

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Asked in :



intuit.

Question 2.

An image is represented by an $m \times n$ integer grid `image` where `image[i][j]` represents the pixel value of the image. You are also given three integers `sr`, `sc`, and `color`. You should perform a flood fill on the image starting from the pixel `image[sr][sc]`.

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Question 3.

Given an $m \times n$ 2D binary grid `grid` which represents a map of '1's (land) and '0's (water), return the number of islands. An island is surrounded by water and is formed by connecting adjacent lands horizontally or vertically. You may assume all four edges of the grid are all surrounded by water.

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Question 4.

Given a reference of a node in a connected undirected graph.

Return a deep copy (clone) of the graph.

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Asked in :



Question 5.

Given an $m \times n$ integers matrix, return the length of the longest increasing path in matrix.

From each cell, you can either move in four directions: left, right, up, or down. You may not move diagonally or move outside the boundary (i.e., wrap-around is not allowed).

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Dynamic Programming



Question 1.

Given an integer array `nums`, find a subarray that has the largest product, and return the product.

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Question 2.

Given a non-empty array `nums` containing only positive integers, find if the array can be partitioned into two subsets such that the sum of elements in both subsets is equal.

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Question 3.

There is a robot on an $m \times n$ grid. The robot is initially located at the top-left corner (i.e., `grid[0][0]`). The robot tries to move to the bottom-right corner (i.e., `grid[m - 1][n - 1]`). The robot can only move either down or right at any point in time.

Given the two integers m and n , return the number of possible unique paths that the robot can take to reach the bottom-right corner.

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